

ModaBot AI

Automated WhatsApp Commerce Engine for High-Volume Retail

Domain: Conversational Commerce / AI Automation	Platform: WhatsApp Business API	Status: Production
--	--	---------------------------

Executive Summary

ModaBot AI was engineered to eliminate high-friction sales drops and delayed response times on WhatsApp for high-volume retail operations. By replacing manual chat operations with a fully automated, event-driven architecture built on the WhatsApp Business Platform, Make, Google Gemini, and integrated database layers, the system handles product inquiries, stock verification, and end-to-end order processing in under two seconds — with zero human intervention on the conversational path.

Metric	Result	Notes
Avg. Response Latency	< 2 seconds	Down from minutes during off-hours
Inquiry Deflection	~70%	Standard stock/price/size queries fully automated
Order Logging Errors	Near-zero	Structured JSON replaces manual spreadsheet entry
Fulfillment Alert Time	Instant	Tri-channel dispatch on order confirmation

Figures reflect typical operating outcomes observed in the automated workflow and are illustrative of expected impact at production scale.

Core Business Bottlenecks & Operational Pain Points

- **High Abandonment Rates:** Delayed responses during off-hours led to lost sales and poor conversion rates.
- **Manual Order Logging:** Human data entry into spreadsheets caused missing addresses, wrong SKU entries, and dispatch delays.
- **Fulfillment Gaps:** Operations teams experienced delays in receiving instant alerts for fast-turnaround orders.
- **Repetitive Catalog Inquiries:** Up to 70% of inbound volume consisted of standard stock, size, and price checks that overburdened support reps.

The Architectural Solution

- **Instant Multi-Turn Conversational Engine:** Integrated Google Gemini to parse customer intent, extract entities (SKUs, quantities, addresses), and strictly contain responses within brand guidelines.
- **Dynamic Inventory Grounding:** Connected Google Sheets directly to Make, allowing the AI to check real-time stock levels, pricing, and variants before confirming orders.

- **Smart Intent-Based Router:** Structured scenario routing to isolate simple product inquiries (INQUIRY) from transaction paths (ORDER).
- **Instant Tri-Channel Dispatch:** Upon capturing an order, the system simultaneously updates the master database, dispatches an HTML alert to the fulfillment team via Gmail, and fires a WhatsApp confirmation receipt to the buyer.

High-Level Data Flow

Step	Stage	Function
1	Inbound WhatsApp Message	Webhook trigger ingests customer message + metadata
2	Gemini NLU Layer	Intent classification + entity extraction (SKU, qty, address)
3	Inventory Grounding	Live lookup against Google Sheets for stock/price/variant validity
4	Router (INQUIRY / ORDER)	Branches conversation into info-response or transaction path
5	Tri-Channel Dispatch	DB write + Gmail alert to ops + WhatsApp confirmation to buyer

Key Technical Features

- **JSON Schema Enforcement:** System prompts force deterministic JSON output (intent, reply, order_data) to prevent parsing failures in downstream modules.
- **Automated Data Validation:** Automatically checks customer phone numbers, item availability, and address completeness prior to execution.
- **Brand-Safe Guardrails:** Hardened system instructions prevent jailbreaks or prompt leakage when users ask out-of-scope technical questions.

Technology Stack

Layer	Technology
Messaging Layer	WhatsApp Business Platform (Cloud API)
Orchestration	Make (event-driven scenario automation)
Conversational AI	Google Gemini (multi-turn NLU + structured generation)
Data Layer	Google Sheets (current) — PostgreSQL / Supabase (roadmap)
Notifications	Gmail (HTML fulfillment alerts), WhatsApp (buyer receipts)

Scalability & Production Roadmap

- **High-Throughput Webhook Handling:** Built on cloud webhooks capable of handling high spikes in concurrent inbound chats without dropping messages.

■ **Enterprise Voice Integration:** Architecture is structured to ingest voice notes by routing audio files through the Whisper API for automated transcription before hitting the LLM.

■ **Direct CRM & ERP Migration:** Flexible JSON schema allows plug-and-play migration from Google Sheets to PostgreSQL, Supabase, Shopify, or custom ERP systems as order volumes exceed spreadsheet limits.

Prepared as a production case study documenting system design, automation logic, and scalability strategy for the ModaBot AI WhatsApp commerce engine.